

M31-2014-DS1 and the Vanishing-Star Problem in CPTG

A Finite Structural-Collapse Interpretation of a Contested Stellar Endpoint

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Abstract

M31-2014-DS1 is one of the best-observed candidates for quiet massive-star disappearance. A hydrogen-depleted supergiant in M31 brightened in the mid-infrared beginning in 2014 and subsequently faded by more than four orders of magnitude in optical light and by more than an order of magnitude in total luminosity. Follow-up JWST and Chandra observations reveal an extremely red remnant, molecular absorption, deep silicate-dust features, a $\sim 40\text{--}200$ au dust structure, $\sim 0.1M_{\odot}$ of molecular gas moving near 100 km s^{-1} , continued bolometric fading to $\log_{10}(L/L_{\odot}) \simeq 3.88$, and no detected X-ray source down to $L_X \lesssim 1.5 \times 10^{35} \text{ erg s}^{-1}$. These observations have produced competing published interpretations: weak mass ejection plus inefficient fallback onto a newly formed black hole, a dust-enshrouded survivor or merger remnant, and a violent binary-interaction alternative.

Curvature Polarization Transport Gravity (CPTG) addresses a different layer of the problem. It retains the GR-controlled exterior spacetime and trapping horizon while restricting the deeper compact continuation to finite-curvature physical states. This paper therefore does not adopt an object-specific mass split or a curvature-threshold classifier and asks the narrower observational question: if M31-2014-DS1 underwent terminal collapse, is its multiwavelength record compatible with a finite CPTG compact endpoint? The answer is affirmative. The luminosity suppression, weak-ejection scale, characterized dusty/molecular environment, optical disappearance, and X-ray nondetection are all compatible with such a channel, but none uniquely determines the hidden interior. M31-2014-DS1 is therefore a high-value, observationally contested candidate for CPTG finite structural collapse and an important population-level counterpart to NGC 6946-BH1.

1 M31-2014-DS1 and the Vanishing-Star Problem

A vanishing-star search reverses the usual transient-discovery problem. Rather than identifying stellar death through a luminous supernova, one begins with a known massive star and asks whether it disappears without producing a normal explosion. This observational strategy was proposed explicitly as a way to search for failed supernovae and direct black-hole formation [1]. M31-2014-DS1 now provides one of the most important examples because the pre-event source, infrared brightening, optical disappearance, and late remnant have all been followed across multiple facilities and wavelengths.

De et al. report a hydrogen-depleted supergiant in M31 that brightened in the mid-infrared beginning in 2014 and then faded dramatically. From 2017 to 2022 the source became undetectable in optical light while also declining strongly in total luminosity. They interpret M31-2014-DS1, together with the earlier NGC 6946 disappearance event, as evidence for failed supernovae forming stellar-mass black holes [2].

The subsequent observational picture is unusually rich. JWST spectroscopy and imaging resolve an extremely red source with strong molecular absorption and deep silicate-dust features, while Chandra provides a stringent X-ray nondetection. De et al. interpret the combined data as a weak ejection of the outer envelope followed by long-lived, low-efficiency fallback onto a newly formed black hole [3]. Beasor et al., analyzing the persistent infrared source, asymmetric dust geometry, and X-ray silence, instead conclude that the fate remains uncertain and that an obscured survivor or stellar-merger interpretation is plausible [4]. Soker further argues that the failed-supernova fallback scenario requires fine tuning and favors a violent binary-interaction interpretation [5].

This disagreement is scientifically useful. The observations establish a genuine observational disappearance event with a strongly altered remnant state, while the physical endpoint remains contested. CPTG enters at that endpoint rather than by replacing the stellar radiation-hydrodynamics. The question addressed here is therefore not whether one can force the light curve into a unique CPTG transient model. It is whether terminal collapse, if that is what occurred, can end in the finite compact solution class admitted by CPTG while remaining consistent with the measured electromagnetic record.

2 CPTG Finite Structural Collapse

CPTG is a nonlinear geometric framework in which ordinary matter sources curvature, curvature develops a polarization response, and organized curvature can be redistributed through transport [8, 9]. In the compact sector, CPTG does not replace General Relativity throughout the black-hole spacetime. GR remains controlling for the exterior geometry, the trapping horizon, orbital dynamics, and the ordinary relativistic strong-field regime. The CPTG compact response is restricted to the deeper high-gradient regime in which the physical solution class excludes an infinite-curvature endpoint.

The retained compact invariant is

$$C_c = g_j g_j + \kappa_c^2 (\partial_i g_j) (\partial_i g_j), \quad (1)$$

with normalized coordinate on the positive structural branch

$$\chi = \frac{C_c}{a_\star^2}, \quad a_\star > 0. \quad (2)$$

Here $a_\star$ is the structural acceleration scale associated with the compact-object solution and κ_c is the compact-sector gradient length. Neither is inferred in this paper from the stellar luminosity, the progenitor mass, or the model-dependent black-hole mass estimate.

The registered high-gradient polarization representative is

$$\mathcal{L}_{\text{pol,c}} = \beta a_\star^{2/3} C_c^{1/3} = \beta a_\star^{4/3} \chi^{1/3}. \quad (3)$$

Here β is the registered compact-sector polarization coefficient. No object-specific numerical value is inferred in this paper.

When the gradient term dominates, the constitutive derivative with respect to the retained field gradient scales as $|\partial g|^{-1/3}$, so the registered high-gradient response softens rather than diverges. Compact transport is likewise required to remain bounded. These conditions restrict the admissible compact solution class, but they do not by themselves supply the complete relativistic interior. For a specified source state and boundary data, the completed strong-field equations must select a finite solution that joins continuously to the GR-controlled exterior.

2.1 Finite Local Center

The local behavior of a regular finite-center branch can be stated without assuming the unknown global profile. Near a regular center, write

$$\Phi(r) = \Phi_0 + \alpha_c r^2 + O(r^4). \quad (4)$$

In local Cartesian coordinates,

$$g_j = -2\alpha_c x_j + O(r^3), \quad (5)$$

so

$$(\partial_i g_j)(\partial_i g_j) = 12\alpha_c^2 + O(r^2). \quad (6)$$

Therefore,

$$C_c(0) = 12\kappa_c^2 \alpha_c^2 < \infty, \quad (7)$$

for finite α_c and κ_c . This establishes the local regularity of the assumed branch. It is not a derivation of the global compact solution; the global finite interior remains a boundary-value problem for the full strong-field equations.

2.2 Relation to Classical Collapse

Black-hole formation is robust in classical GR. The distinction relevant here is the completion of the trapped interior, not whether a trapping horizon can form. Under the standard singularity-theorem hypotheses, trapped collapse leads to geodesic incompleteness rather than to a derived finite material interior [7]. CPTG retains the successful GR exterior while requiring the deeper physical continuation to remain finite-curvature. The observational task is therefore first to establish whether a star has undergone terminal collapse; only then does the CPTG finite-continuation question become physically applicable.

3 Observed State of M31-2014-DS1

The event is best treated as a chronology of measured state changes rather than as a single inferred remnant mass. The Science discovery paper establishes the pre-event source, the 2014 mid-infrared brightening, the subsequent optical disappearance, and the large decrease in total light [2]. The later JWST and Chandra observations add a characterized red infrared remnant, molecular absorption, dust structure, gas kinematics, continued bolometric fading, and an X-ray upper limit [3]. Beasor et al. recover the persistent infrared source but emphasize asymmetric dust and the possibility that its inferred luminosity is only a lower limit [4].

Table 1: Observed and literature-derived quantities defining the current state of M31-2014-DS1. Model-dependent quantities are identified explicitly and are not adopted as CPTG compact-state measurements.

Quantity	Published value/state	Role in this paper
Object	M31-2014-DS1	Vanishing-star / failed-supernova candidate in M31.
Progenitor	Hydrogen-depleted supergiant; De et al. model $\sim 12\text{--}13M_{\odot}$	Establishes a massive stellar source; mass is model context rather than the CPTG classifier.
2014 event	Mid-infrared brightening without a normal luminous supernova	Defines the weak-transient / disappearance channel.
Optical evolution	Faded by $\gtrsim 10^4$ and became undetectable	Direct disappearance observable.
Total luminosity	Faded by $\gtrsim 10$ from the pre-event state	Establishes large bolometric suppression.
JWST remnant	Extremely red source with molecular absorption and deep silicate-dust features	Confirms a persistent obscured remnant environment.
De et al. luminosity	$\log_{10}(L/L_{\odot}) \simeq 3.88$, about 7–8% of progenitor	Quantitative fading branch used in Sec. 4.
Beasor et al. luminosity interpretation	Current source roughly 7–25% of progenitor; inferred luminosity may be a lower limit	Encodes the asymmetric-dust systematic and survivor/merger alternative.
Dust structure	Approximately 40–200 au	Geometric scale of the circumstellar remnant environment.
Molecular gas	$\sim 0.1M_{\odot}$ moving at $\sim 100\text{ km s}^{-1}$	Sets the observed/modelled weak-ejection energy scale.

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Quantity	Published value/state	Role in this paper
X-ray state	No detected source; 3σ limit $L_X \lesssim 1.5 \times 10^{35} \text{ erg s}^{-1}$ in 0.5–7 keV	Constraint on exposed accretion and remnant energetics under the published spectral/absorption assumptions.
Conventional BH estimate	$\sim 5M_\odot$ in the De et al. failed-SN model	Model-derived endpoint context; not a dynamical mass measurement and not used as a CPTG compact-state input.

The observed sequence can therefore be summarized as

$$\begin{aligned}
& \text{mid-infrared brightening} \\
& \quad \downarrow \\
& \text{optical disappearance} \\
& \quad \downarrow \\
& \text{strong bolometric fading} \\
& \quad \downarrow \\
& \text{dust + molecular remnant} \\
& \quad \downarrow \\
& \text{X-ray nondetection.}
\end{aligned} \tag{8}$$

This sequence is the observational basis of the paper. It does not require assigning the remnant a CPTG core mass, polarization amplification, curvature threshold, or transport amplitude.

4 Bolometric Fading and the Persistent Infrared Source

De et al. obtain a late luminosity

$$\log_{10} \left(\frac{L_{\text{rem}}}{L_\odot} \right) \simeq 3.88, \tag{9}$$

which corresponds to

$$L_{\text{rem}} \simeq 7.59 \times 10^3 L_\odot. \tag{10}$$

Using their approximate progenitor luminosity scale $\log_{10}(L_{\text{prog}}/L_\odot) \simeq 5$, the central luminosity ratio is

$$R_L^{\text{De}} = \frac{L_{\text{rem}}}{L_{\text{prog}}} \simeq 10^{-1.12} = 0.0759. \tag{11}$$

Thus the De et al. branch places the late source at about 7–8% of the progenitor luminosity [3].

The physical interpretation of that luminosity is disputed. Beasor et al. also find a strongly diminished source, but their asymmetric dust model implies that the observed reprocessed luminosity may be only a lower limit because radiation can escape preferentially away from the line of sight. They estimate a present source at roughly 7–25% of the progenitor luminosity and argue that a dust-enshrouded survivor or merger remains viable [4]. The robust conclusion is therefore not one unique value of R_L , but substantial long-term suppression together with a persistent dusty infrared source.

That distinction is important for CPTG. Luminosity suppression is evidence about the stellar endpoint and remnant environment; it is not a direct measurement of the hidden compact geometry. If a normal stellar photosphere eventually re-emerges as the dust clears, the terminal-collapse interpretation would be strongly weakened. Continued absence of such a survivor while the source remains substantially below the progenitor luminosity would strengthen the collapse interpretation.

5 Weak Ejection, Molecular Gas, and Dust Geometry

The JWST analysis gives a molecular-gas scale near

$$M_{\text{gas}} \sim 0.1M_{\odot}, \quad v_{\text{gas}} \sim 100 \text{ km s}^{-1}. \quad (12)$$

The corresponding kinetic-energy scale is

$$E_{\text{kin}} = \frac{1}{2}M_{\text{gas}}v_{\text{gas}}^2 \simeq 9.94 \times 10^{45} \text{ erg}, \quad (13)$$

consistent with the published characterization of an approximately 10^{46} erg low-energy ejection [3]. The measured/modelled gas mass and velocity therefore recover the weak-ejection scale directly from the remnant kinematics.

The dust continuum is modeled over a characteristic radial range of approximately

$$40 \text{ au} \lesssim R_{\text{dust}} \lesssim 200 \text{ au}. \quad (14)$$

A parcel moving at 100 km s^{-1} for ten years travels

$$R_{\text{travel}} \simeq 2.11 \times 10^2 \text{ au}, \quad (15)$$

which is of the same order as the inferred dust structure. The comparison supports the dynamical plausibility of a decade-old weak ejecta/remnant environment, but it does not identify the geometry as curvature transport. In this paper the dust and gas are astrophysical tracers of the post-event environment, not direct measurements of a CPTG transport field.

6 X-ray Nondetection

The 2024 Chandra observations yield a 3σ upper limit in the 0.5–7 keV band,

$$L_X \lesssim 1.5 \times 10^{35} \text{ erg s}^{-1}. \quad (16)$$

De et al. convert the count-rate limit assuming a power-law photon spectrum with $\Gamma = 2$ and Galactic absorption corresponding to $N_H = 2.26 \times 10^{21} \text{ cm}^{-2}$ [3]. The numerical luminosity limit should therefore be read with those spectral and absorption assumptions.

Using Eq. (10) and $L_\odot = 3.828 \times 10^{33} \text{ erg s}^{-1}$, the De et al. bolometric branch gives

$$L_{\text{bol}} \simeq 2.90 \times 10^{37} \text{ erg s}^{-1}, \quad (17)$$

and hence

$$\frac{L_X}{L_{\text{bol}}} \lesssim 5.2 \times 10^{-3}. \quad (18)$$

The nondetection is an important remnant constraint, but the 2026 literature demonstrates that it is not presently a unique classifier. De et al. accommodate the X-ray silence through very low-efficiency fallback and obscuration [3]; Beasor et al. argue that the lack of X-rays disfavors accretion-powered late-time luminosity and is compatible with an obscured stellar survivor or merger [4]. CPTG does not require X-ray silence as a defining property of a finite compact interior. Accretion and radiative transfer occur outside the hidden compact continuation and can vary independently. The X-ray upper limit should therefore be retained as a strong environmental constraint rather than promoted to a CPTG discriminator.

7 Competing Interpretations of the Event

7.1 Failed Supernova with Weak Ejection and Fallback

The De et al. interpretation is a coherent conventional collapse model. A hydrogen-depleted progenitor undergoes core collapse, a weak shock ejects roughly the outer hydrogen-rich material, most of the star forms a black hole, and a small amount of loosely bound matter powers long-term inefficient fallback. The measured gas scale, dust environment, fading luminosity, and X-ray nondetection are fitted within that picture [2, 3]. The endpoint is a conventional stellar-mass black hole; the weak transient and remnant emission arise from stellar-collapse and accretion physics rather than from GR alone.

7.2 Dust-Enshrouded Survivor or Merger

Beasor et al. emphasize that a luminous infrared source remains at the position approximately a decade after the event, that the dust geometry is strongly asymmetric, and that

the bolometric luminosity inferred from a spherical reprocessing model can underestimate the true central luminosity. They therefore regard a surviving dust-enshrouded star or merger remnant as plausible and identify future re-brightening after dust clearing as a decisive possibility [4].

7.3 Violent Binary Interaction

Soker argues that the failed-supernova fallback construction requires a finely tuned combination in which only a very small fraction of bound material is accreted while outflow feedback fails to terminate fallback for more than a decade. He also argues that the interaction of the outflow with surrounding material should radiate more strongly than observed and favors a violent binary-interaction / intermediate-luminosity transient interpretation [5]. Whether or not that alternative ultimately succeeds, it reinforces the central observational fact: the endpoint is not yet uniquely identified.

Table 2: Interpretive comparison for M31-2014-DS1. CPTG differs most directly at the hidden compact endpoint; it is not used here as a replacement for the stellar radiative-transfer or ejecta calculations.

Observable / question	Failed SN + fallback	Survivor / merger / binary interaction	CPTG finite structural collapse
Optical disappearance	Terminal stellar collapse	Dust obscuration after merger or violent interaction	Compatible if terminal collapse occurred
Persistent MIR source	Dust plus fading fallback-powered central source	Obscured surviving star / merger remnant	Environmental tracer; not itself a compact-state observable
Weak ejecta	Low-energy envelope ejection	Binary-interaction ejecta can also be weak	Compatible external phenomenology; not uniquely predicted by the finite interior
X-ray nondetection	Very inefficient and/or obscured accretion	No exposed accreting BH required	Constraint on exterior accretion; not a finite-interior discriminator
Hidden endpoint	Stellar-mass classical black hole	Surviving star or merger product	Finite-curvature compact continuation behind a GR-controlled exterior, if terminal collapse is confirmed

8 What CPTG Adds

The observational dispute separates naturally into two layers. The first is stellar-collapse phenomenology: did the star actually undergo terminal collapse, or is the infrared source a surviving merger product hidden by asymmetric dust? The second exists only if terminal collapse occurred: what is the physical completion of the trapped compact interior?

CPTG changes the second layer. The $\sim 5M_{\odot}$ black-hole mass in the failed-supernova interpretation is a model-derived endpoint estimate rather than an exterior dynamical mass measurement, so it is not used here as a CPTG apparent-mass input. The present data likewise do not select a numerical compact-core mass. No M_{core} , A_{pol} , χ , or compact

transport amplitude is inferred from M31-2014-DS1 in this paper; those quantities belong to the strong-field solution and cannot be reconstructed uniquely from the present electromagnetic data.

What CPTG supplies is instead a definite physical admissibility statement: a terminally collapsed object may possess the same GR-controlled trapping exterior while the deeper continuation belongs to a finite-curvature compact solution rather than an infinite-curvature physical endpoint. M31-2014-DS1 becomes a CPTG target precisely because its electromagnetic state is compatible with terminal disappearance, not because the dust, molecular gas, or X-ray limit directly measure the interior geometry.

This interpretation is strongest if the central star never returns. If future observations recover a normal stellar photosphere at a luminosity compatible with a surviving system, the terminal structural-collapse classification fails. If the source continues to fade, remains optically absent as the dust evolves, and increasingly excludes a normal survivor, the finite-collapse interpretation becomes correspondingly stronger.

9 M31-2014-DS1 and NGC 6946-BH1 as a Population Test

M31-2014-DS1 should not be interpreted in isolation. De et al. explicitly connect it with the earlier NGC 6946-BH1 disappearance and interpret the two events as evidence for failed supernovae forming stellar-mass black holes [2]. Recent analysis likewise treats NGC 6946-BH1 and M31-2014-DS1 as two promising failed-supernova candidates and finds a striking population-level luminosity contrast: late merger remnants in their comparison sample are typically 10–100 times more luminous than their progenitors, whereas the two failed-supernova candidates are roughly an order of magnitude dimmer [6].

The two objects are complementary rather than redundant. NGC 6946-BH1 provides a long-baseline optical disappearance with a strongly suppressed dusty remnant; M31-2014-DS1 provides unusually detailed molecular, dust, and X-ray information but currently carries a sharper published debate over survival versus collapse. If both are ultimately established as terminal-disappearance events, the pair begins to define a repeatable observational channel rather than a one-object anomaly. If M31-2014-DS1 re-emerges as a survivor while NGC 6946-BH1 does not, the contrast will help identify which observables actually separate terminal collapse from merger-like impostors.

For CPTG the population question is therefore concrete: after survivor, merger, and obscuration channels are removed, does a recurrent class of massive stars remain that disappear into long-lived compact-remnant states without a normal supernova? A positive answer would establish the astrophysical channel in which the CPTG finite compact continuation can be tested. It would not, by itself, determine the hidden strong-field solution.

10 Discriminating Observations and Falsifiability

The near-term discriminant is the fate of the central luminous source. Recovery of a normal optical or near-infrared stellar photosphere as the dust column declines, especially at a luminosity approaching the progenitor state, would strongly favor survival or merger and would remove M31-2014-DS1 from the terminal CPTG structural-collapse class. Continued optical absence accompanied by declining or cooling infrared emission would instead strengthen the terminal-collapse interpretation.

The X-ray evolution constrains fallback and accretion but is not a direct test of the finite CPTG interior. Emergence of an accretion-powered X-ray source as obscuring material thins would support the presence of a compact accretor; continued X-ray silence would constrain the accretion efficiency and geometry. Either outcome can occur outside a finite compact continuation, so the X-ray test is primarily astrophysical rather than uniquely geometric.

A genuinely CPTG-specific strong-field test ultimately requires an observable sensitive to the compact continuation itself rather than only to the obscuring envelope. The present electromagnetic record can establish or reject M31-2014-DS1 as a terminal-collapse candidate. It cannot, by itself, reconstruct the unique interior geometry. That boundary makes the test program cleaner: first determine whether the star survived; if it did not, the object becomes a direct target for future compact-remnant probes.

11 Conclusion

M31-2014-DS1 is scientifically important because the disappearance is real while the endpoint remains contested. The source brightened in the mid-infrared, became undetectable optically, faded strongly in bolometric luminosity, and now appears as an extremely red dust- and molecule-rich remnant with a stringent X-ray upper limit. The inferred gas mass and speed independently recover a weak-ejection energy near 10^{46} erg, and the decade-scale travel distance is comparable to the inferred dust-structure scale.

The 2026 literature provides three serious readings of that same record. De et al. interpret it as weak ejection and inefficient fallback during black-hole formation. Beasor et al. retain a dust-enshrouded survivor or merger because the central luminosity can be underestimated by asymmetric dust. Soker argues for a violent binary-interaction alternative. The correct publication-level statement is therefore not that the event has already selected one endpoint, but that it has become an unusually discriminating vanishing-star experiment.

CPTG contributes a distinct physical possibility if terminal collapse is confirmed. GR continues to control the exterior spacetime and trapping horizon, while the deeper CPTG solution class excludes an infinite-curvature physical endpoint and admits a finite structural continuation. No object-specific CPTG mass split or curvature threshold is needed to make that statement, and none is inferred here from the current data.

The result of this paper is therefore affirmative and testable: *M31-2014-DS1 is a high-value, observationally contested candidate for CPTG finite structural collapse.* A recovered

normal stellar survivor would remove it from that class. Continued disappearance and fading would strengthen the terminal-collapse interpretation and, together with NGC 6946-BH1, would sharpen the population-level test of finite compact-state formation.

Data, Code, and Evidence Availability

The observational quantities used in this paper are taken from the cited Science, Astrophysical Journal Letters, Monthly Notices of the Royal Astronomical Society, and Open Journal of Astrophysics studies. The JWST observations analyzed by De et al. are associated with JWST program 6809 and MAST DOI 10.17909/c3sv-qh71; the Chandra data collection is associated with DOI 10.25574/cdc.538 [3]. No compact-remnant mass, CPTG core mass, polarization amplification, curvature threshold, or transport amplitude is fitted or inferred in the present analysis. The luminosity ratio, kinetic-energy scale, travel-distance scale, and X-ray-to-bolometric ratio are direct algebraic combinations of the cited observational/model quantities. CPTG theory papers and supporting public research resources are maintained through the project repository [10].

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